

Files, Terminals, and Your First Python Program

CS1 Week 3 — Monday, August 31, 2026

Computer Science I

The lesson in one sentence

The file manager and terminal are two views of the same filesystem; the computer follows exact instructions.

Predict. Try one small change. Read the result. Verify.

Two views, one location

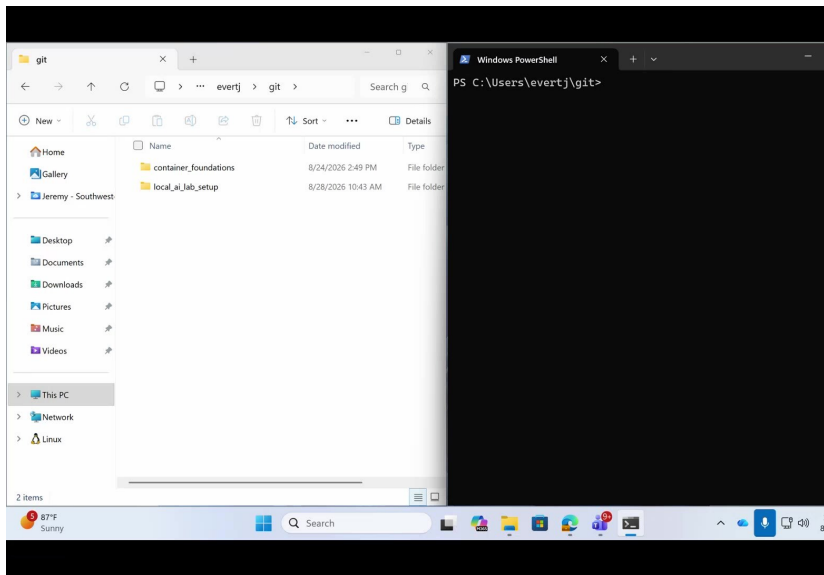
File manager

- Visual folders and files
- Create and open items
- Save changes explicitly

Terminal

- Text commands
- Exact current location
- Ask it to list or inspect

From class: the same git folder, in the file manager and in PowerShell

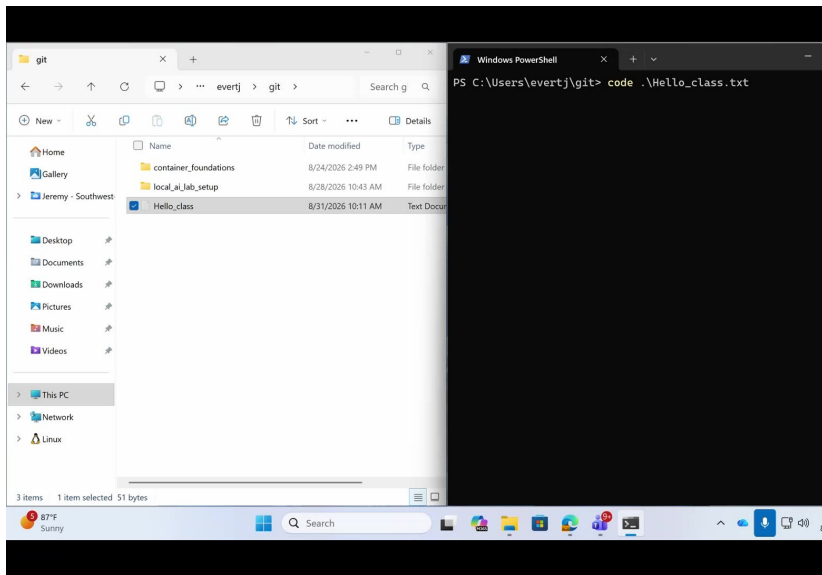


Where am I? What is here?

```
ls  
cd GIT
```

- `ls` lists the current directory.
- `cd` changes the current directory.
- After creating a file, list again: the terminal will not guess.

From class: Hello_class.txt now shows up in the folder after we saved it



Files have jobs and names

Extension	Kind	Use
.txt	plain text	notes or simple text
.md	Markdown	structured readable notes
.py	Python source	instructions Python can execute

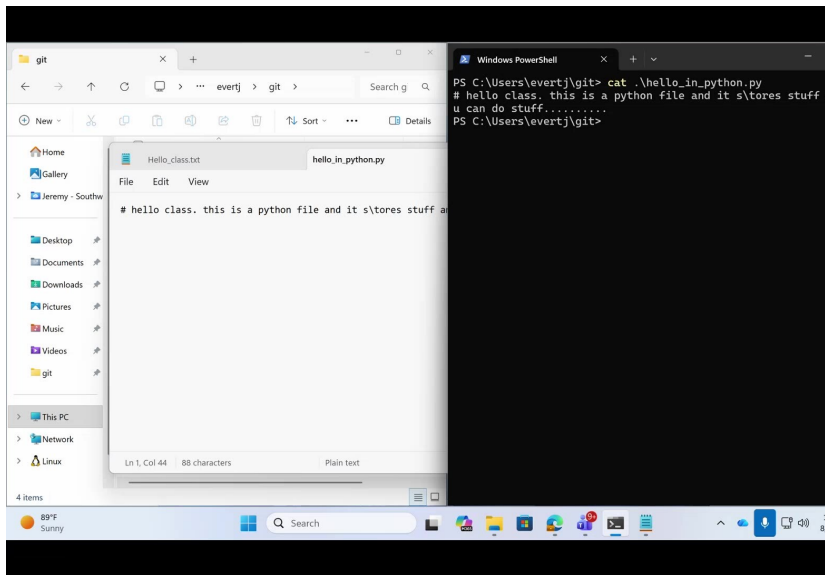
A .py file is text on disk; it becomes a program when its contents and execution context are correct.

Your first tiny Python file

```
# A short note for another human  
print("hello world")
```

- # begins a comment: useful to humans, ignored by Python.
- print() is a function that sends text to output.

From class: a real comment line inside the file, read back with cat



From class: naming the parts: the function, the argument, the string

The screenshot shows the Microsoft Copilot web interface. The browser tabs include 'QuickLaunchSSO', 'Python Print Function Case', 'LinkedIn Learning: Online C...', 'Dashboard', and 'Microsoft Education'. The URL bar shows a Microsoft chat conversation link.

On the left sidebar, the 'Copilot' section is active, with 'Chat' selected over 'Cowork'. Below this are links for 'New chat', 'Search', 'Library', 'Teach', and 'Notebooks'. Under the 'Agents' section, there are icons for 'Researcher', 'Analyst', 'Excel', 'PowerPoint', and 'New agent'. At the bottom of the sidebar, there's a 'Work' tab with 'Evert, M365 Copilot (Premium)' selected.

The main chat area has a header with 'Work IQ' and 'GPT 5.6 Think'. Below this is a large text input field. The chat history shows two messages:

5. In the working line below, identify:

- the function name
- the argument
- the string

Below the list is a code block containing the following Python code:

```
1 print('Hello, World!')
```

Below the code block is a question:

6. Bonus prediction: What would this print?

At the bottom of the chat area is a 'Message Copilot' input field with a microphone icon. Below the input field, a small disclaimer reads: 'AI-generated content may be incorrect'.

The Windows taskbar at the bottom shows the date and time as 89°F Sunny, a search bar, and several application icons including File Explorer, Microsoft Edge, and the Copilot app.

Literal machine, useful partner

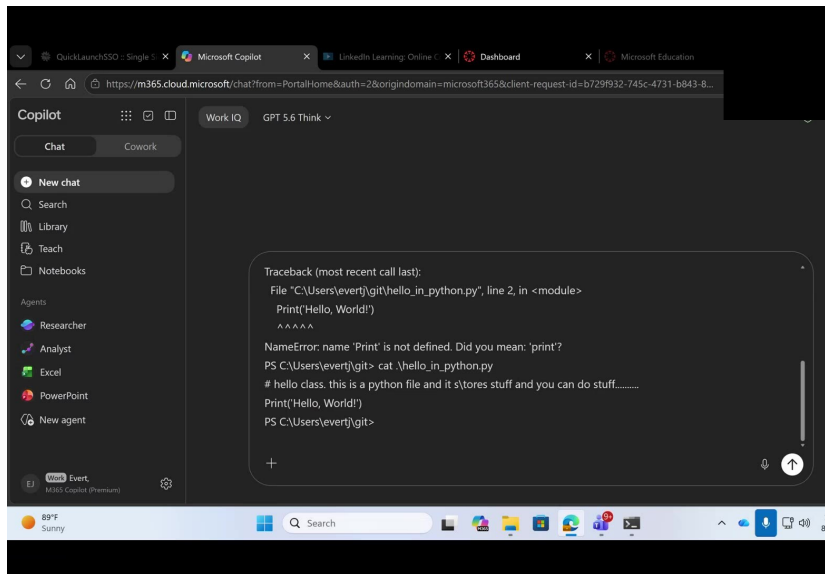
- Python does not infer your intention.
- Names, punctuation, capitalization, and location matter.
- Once you learn the language, the machine becomes a resource instead of a mystery.

Case sensitivity is a rule

```
print("hello world")    # works  
Print("hello world")    # different name: error
```

Humans see the intended word. Python sees two different spellings.

From class: Print vs print — “name 'Print' is not defined. Did you mean: 'print'?”



Errors are information

- ➊ Read the error and locate the line.
- ➋ Predict one small correction.
- ➌ Change one thing and run again.
- ➍ Ask a person or tool to translate unfamiliar language.

An error costs a little time; the correction teaches you what the rule is.

From class: one small fix, then run it: python
.\hello_in_python.py

The screenshot shows a Windows desktop with a dark theme. On the left is the Microsoft Copilot sidebar, which includes a 'Chat' tab, a 'COWORK' section, and a list of agents: Researcher, Analyst, Excel, PowerPoint, and a 'New agent' button. The main area of the Copilot window displays a chat conversation with 'GPT 5.6 Think'. The chat history shows a prompt to 'Fix it with one cop' and a response that 'This replaces the entire file,' followed by a code block. The code block contains a Python script with a docstring, a comment, a print statement, and a command to set the content of a file. The chat also shows the 'Expected ending:' and a 'Message Copilot' button.

On the right, a Windows PowerShell terminal window is open, showing the execution of the Python script. The terminal output is as follows:

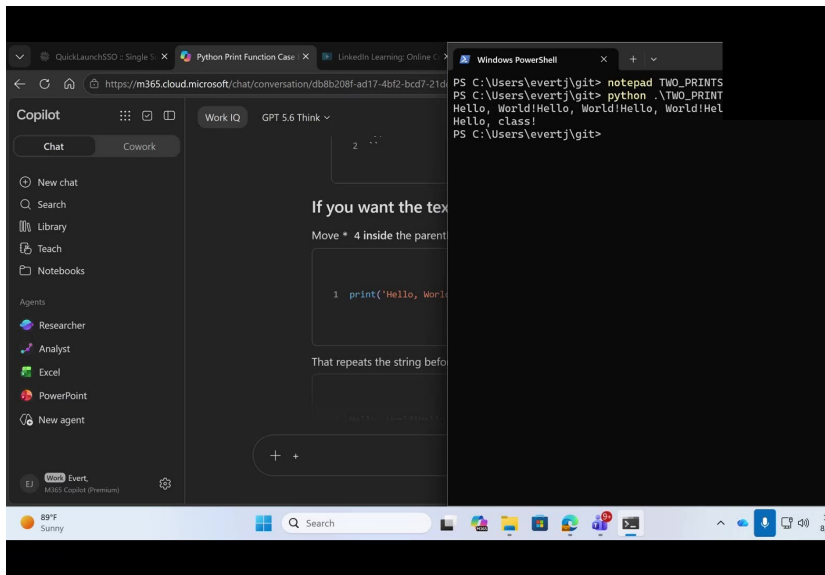
```
PS C:\Users\eventj\git> @'  
>> # Hello, class. This is a Python file.  
that Python can execute.  
>> print('Hello, World!')  
>> '@ | Set-Content -Path .\hello_in_python.py -Encoding u  
PS C:\Users\eventj\git>  
PS C:\Users\eventj\git> Get-Content .\hello_in_python.py  
# Hello, class. This is a Python file. It stores instructi  
at Python can execute.  
print('Hello, World!')  
PS C:\Users\eventj\git> python .\hello_in_python.py
```

The Windows taskbar at the bottom shows the date and time as 8:15, the weather as 89°F Sunny, and various system icons.

Small tools from the lecture

- **Up arrow:** recall the previous terminal command.
- **Tab:** complete a name and check available choices.
- **Ctrl+S:** save the file before testing it.
- **File inspection:** look at the text before asking what it does.

From class: experimenting: TWO_PRINTS.py and the repeated output it makes



Before Wednesday

- Save a tiny `.py` file with one comment and one `print()` line.
- Predict its output before running it.
- Wednesday: paired reasoning with the Week 3 material.
- Friday: show-and-tell/check-in on what worked and what you learned.

Be precise, be curious, and use errors as evidence.